Name: ____________________________ Date: ________________ Period: _________

Spiral Review — Key

1. Complete the table of exact values.

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\cos t$	1	0	-1	0	1
$\sin t$	0	1	0	-1	0

2. The graph of $y = \sin(x)$ is shifted **up 3** units and **scaled vertically by 2**. Write the equation of the transformed function.

$y = 2\sin(x) + 3$

3. A point starts at (1, 0) and travels *counterclockwise* around a circle of radius 1. Fill in the table.

Turn	**New coordinates**
After a quarter turn ($\frac{1}{4}$ of the way around)	(0, 1)
After a half turn ($\frac{1}{2}$ of the way around)	(−1, 0)
After a full turn (all the way around)	(1, 0)

Teacher Note

Q1 activates the cosine and sine values students need for the unit-circle parametrization. Q2 previews the transformation template $(h + r\cos t,\ k + r\sin t)$ — if students struggle here, connect the vertical shift to the center height in the Ferris wheel problem. Q3 is a geometric preview of EK 4.4.A.1; connect to the table in the notes.